Oswin Franco Cervantes

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Technical Profile

Electrical engineering and physics student focused on analog/mixed-signal ICs, RF/microwave systems, and applied electromagnetics; experienced in device modeling, simulation, layout, fabrication, and laboratory validation.

Education

Massachusetts Institute of Technology, Cambridge, MA — **Expected May 2028**
B.S. Candidate in Physics and Electrical Engineering with Computing — Sept. 2024–Present
Relevant coursework: **MIT EE:** High-Frequency Integrated Circuits (graduate), Electromagnetics (graduate), Solid-State Circuits, Semiconductor Electronic Circuits, Signal Processing, Digital Systems Laboratory (Verilog); **MIT Physics:** Quantum Physics I, Statistical Mechanics, Classical Mechanics II, Vibrations and Waves; **Professional:** IEEE MTT-S IMS2026 RF Boot Camp; **Coursera:** Microwave Engineering and Antennas, RF and Millimeter-Wave Circuit Design

Georgia Tech, Atlanta, GA — Dual Enrollment, Mathematics/Computer Science; GPA: 4.0/4.0 — **Aug. 2022–May 2024**

Selected Analog IC Design Project

22 nm FinFET Amplifier and Control Chip — **Spring 2026**
MIT Semiconductor Electronic Circuits — *Cambridge, MA*

- Designed and simulated a 22 nm FinFET CMOS analog amplifier with activation switches and digital control, applying semiconductor device physics and large- and small-signal models.
- Completed the Cadence schematic and physical layout for a course tapeout; used Calibre DRC/LVS to verify fabrication rules and schematic-layout consistency.
- Verified in simulation that the amplifier exceeded 26 dB gain at 1 MHz and 300 mV peak-to-peak output-swing targets while operating 48% below its 500 μW power limit.

Relevant Experience

Undergraduate Researcher — Prof. William D. Oliver — **Sept. 2025–Present**
MIT Research Laboratory of Electronics, Engineering Quantum Systems (EQuS) Group — *Cambridge, MA*

- Design and test niobium coplanar-waveguide LC filters for hybrid quantum hardware; benchmark 2nd-, 4th-, and 8th-order designs with a 100 MHz cutoff and minimal passband insertion loss, with the 8th-order filter achieving 35.91 dB worst-case, 48.52 dB median, and 60.77 dB peak attenuation across the 3–5 GHz stopband.
- Create RF layouts in KLayout and model eigenmodes and S-parameters in Ansys HFSS; automate simulations with PyAEDT, extract parasitics in Q3D/Maxwell, and develop a machine-learning model to optimize higher-order LC-filter parameters.
- Develop reusable components and fabrication-ready GDS layouts in CDES, a Python/gdstk superconducting-circuit framework, using GitHub version control; fabricate niobium-on-silicon devices through cleanroom photolithography.

Hardware Development Engineer (Systems Engineering) Intern — **May 2025–Aug. 2025**
Amazon Lab126, Advanced Products — *Sunnyvale, CA*

- Designed and simulated buck, boost, and buck-boost power-regulation circuits in OrCAD Capture and SPICE; laid out PCBs in Cadence Allegro and evaluated efficiency and maximum power point tracking (MPPT) with oscilloscopes and electronic loads.
- Worked cross-functionally to translate ambiguous product requirements into system architecture, block diagrams, schematics, and PCB layout for a multi-IC solar-charging module compatible with Amazon Kindle devices.
- Integrated and debugged ICs over I^2C; assembled and tested 0402, 0603, and 0804 SMD prototypes and debug boards to validate system performance; trained new hires on PCB tools.

Avionics Engineer — **Jan. 2026–Present**
MIT Rocket Team — *Cambridge, MA*

- Design RF circuits and PCB interconnects in Altium; document electronic parts and specifications; validate HFSS Driven Terminal and SIwave S-parameter models against vector network analyzer measurements.
- Reduce RF component cost by designing custom 5.8 GHz and 915 MHz microstrip quadrature couplers on FR-4 PCBs to replace expensive stripline parts and unsuitable LTCC alternatives.

Technical Skills

Analog/RF Design: 22 nm FinFET CMOS, analog and mixed-signal ICs, solid-state circuits, semiconductor device modeling, large- and small-signal analysis, power conversion, RF and microwave circuits, S-parameters, transmission lines, impedance matching, filters and resonators, SPICE, PCB design, digital and RTL design
Layout, Fabrication, and Test: Cadence Virtuoso/Allegro/AWR Microwave Office, Ansys HFSS/Maxwell/Q3D/SIwave/PyAEDT, Calibre DRC/LVS, Altium, OrCAD, KLayout; GDS layout, niobium-on-silicon photolithography, cleanroom processing, VNA, oscilloscopes, electronic loads, SMD assembly
Programming/HDL: Python, MATLAB, C, Java, Julia, Bash, Verilog, VHDL, Bluespec, RISC-V assembly; NumPy, pandas, gdstk, Git

Honors and Certification

Honors: QuestBridge Scholarship; Amazon Scholar; Jack Kent Cooke Foundation Semifinalist; Stanford Intrologic Scholar
Certification: TestOut Network Pro Certification